

Glutathione

Master Antioxidant | 600mg & 1,500mg x 10 Vials

IV / IM / SubQ | Skin Brightening | Immune & Liver Support

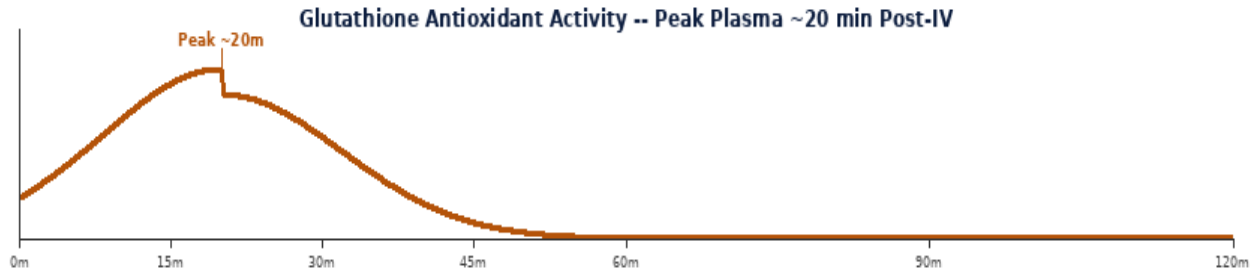
Purity: 99.8% | Endotoxin-Free | Research Use Only

Glutathione (GSH) is the most abundant endogenous antioxidant tripeptide (gamma-Glu-Cys-Gly). It is central to cellular redox homeostasis, xenobiotic detoxification, immune function, and melanogenesis inhibition. Injectable glutathione achieves plasma concentrations far exceeding oral supplementation, which has poor bioavailability due to gut breakdown. Available in 600mg (standard IV) and 1,500mg (high-dose) vials.



Parameter	Specification
Class	Endogenous tripeptide antioxidant -- GSH (gamma-Glu-Cys-Gly)
Vial Sizes	600mg x 10 1,500mg x 10 Purity 99.8%
Reconstitution	600mg: 10mL saline or BAC Water -> 60mg/mL 1,500mg: 10mL -> 150mg/mL
Standard Dose	600-1,500mg per session IV push / IV drip / IM
Route	IV push (2-3 min) IV drip (20-30 min) IM SubQ (lower dose)
Frequency	2-3x per week IV/IM daily SubQ lower dose no fasting required
Cycle & Storage	Ongoing with monitoring 2-8 deg C use within 24 hr reconstituted

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
Antioxidant & Redox Control	GSH directly neutralises reactive oxygen species (ROS) and reactive nitrogen species; regenerates vitamins C and E; cofactor for glutathione peroxidase (GPx); central to cellular redox balance and protection from oxidative stress	Cellular protection; reduced oxidative damage; anti-aging research
Liver Detoxification	Conjugates with toxins and drugs via glutathione-S-transferase (GST); essential for phase II liver detoxification; protects hepatocytes from oxidative damage; supports liver regeneration	Hepatoprotection; detoxification; liver health research
Melanogenesis Inhibition	Inhibits tyrosinase enzyme (melanin synthesis rate-limiting step); shifts eumelanin -> pheomelanin pathway; reduces melanin production in melanocytes; systemic skin brightening effect	Skin brightening; hyperpigmentation research; melanogenesis inhibition

Benefit	Context
Master Antioxidant	GSH is the primary intracellular antioxidant -- it regenerates other antioxidants, protects DNA, and maintains cellular redox balance.
Liver & Detox	Central to hepatic phase II detoxification. IV glutathione is used clinically for acetaminophen overdose and liver protection protocols.
Skin Brightening	Inhibits tyrosinase and shifts melanin synthesis pathway. Systemic IV/IM injection produces progressive skin brightening over weeks.

RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



1mL Insulin Syringe | 29-31G | SubQ

Formula: $\text{Vol(mL)} = \frac{\text{Dose(mcg)}}{\text{Conc(mcg/mL)}}$ | **Units:** $\text{Vol} \times 100$

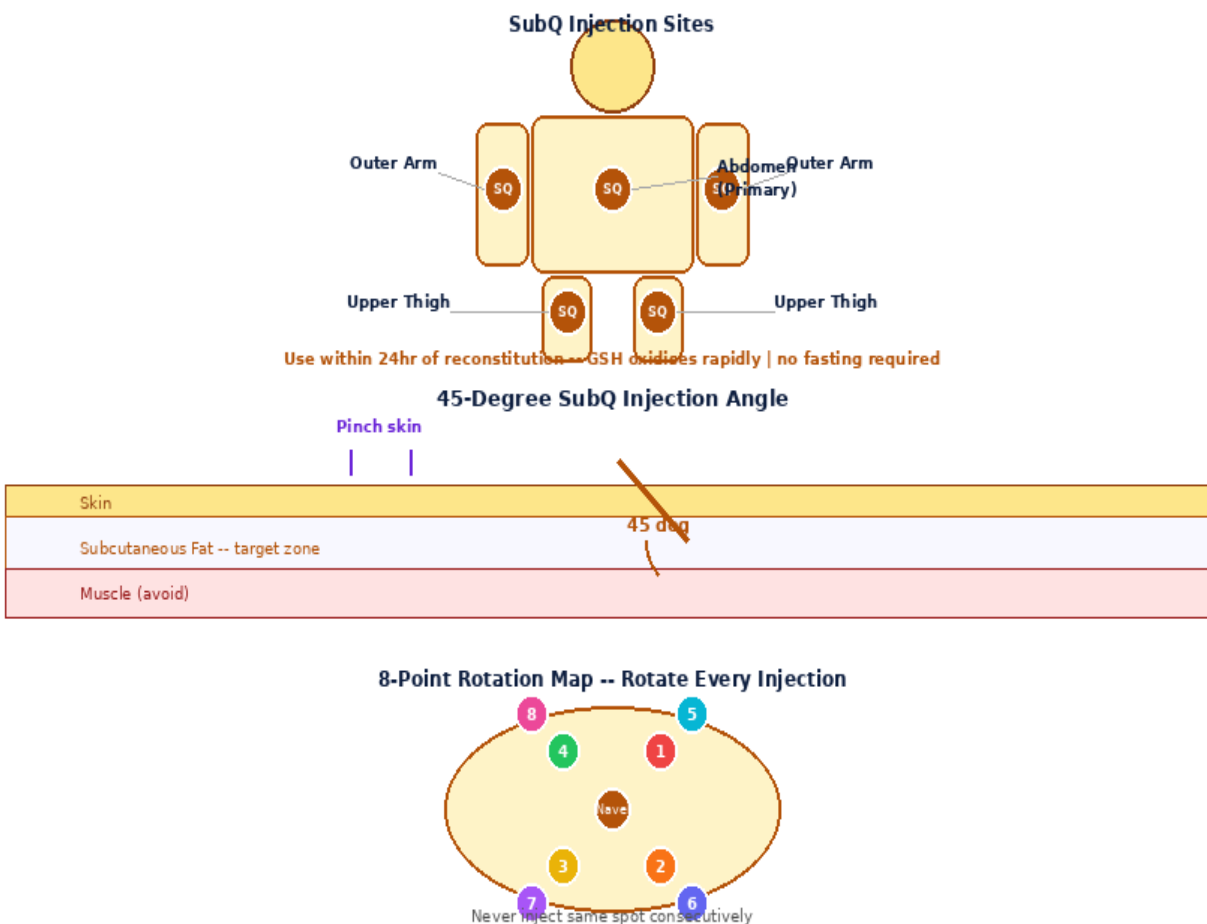
600mg x 10 -- 10mL BAC -> 60.0mg/mL (60,000mcg/mL)

Dose(mcg)	600mcg	1000mcg	1500mcg	2000mcg	3000mcg
Vol(mL)	0.0100	0.0167	0.0250	0.0333	0.0500
Units	1.0U	1.7U	2.5U	3.3U	5.0U
Doses/Vial	100	60	40	30	20

1500mg x 10 -- 10mL BAC -> 150.0mg/mL (150,000mcg/mL)

Dose(mcg)	600mcg	1000mcg	1500mcg	2000mcg	3000mcg
Vol(mL)	0.0040	0.0067	0.0100	0.0133	0.0200
Units	0.4U	0.7U	1.0U	1.3U	2.0U
Doses/Vial	250	150	100	75	50

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



Step	Action	Detail
1	Reconstitute	Add 10mL saline or BAC Water to vial. Swirl gently. Clear solution required.
2	Select route	IV push: inject over 2-3 min. IV drip: dilute in 100-250mL saline, infuse 20-30 min. IM: draw into syringe, gluteal or deltoid. SubQ: abdomen, lower dose.
3	Administer	IV: slow push or drip. IM: 23-25G x 1-1.5in. SubQ: 29-31G x 5/16in.
4	Use immediately	Reconstituted GSH oxidises rapidly. Use within 24 hours. Discard remainder.

Protocol	Dose	Frequency	Duration	Notes
Standard IV/IM	600-1,000mg	2-3x/week	Ongoing w/ monitoring	IV push 2-3min or IM gluteal.
High-Dose IV	1,500mg	2-3x/week	Ongoing w/ monitoring	IV drip over 20-30 min. Clinical setting preferred.

Side Effect	Likelihood	Management
Flushing / warmth (IV)	Occasional	Rate-related. Slow infusion rate.
Nausea (IV push)	Occasional	Slow push rate over 2-3 min minimum.
Skin lightening (systemic)	With continued use	Expected melanogenesis effect. Monitor and adjust.

Asthma aggravation	Rare	Sulphur-containing. Avoid in sulphite-sensitive asthma.
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Contraindications: Cancer history, pregnancy, under 18.

1. 600mg vial: 10mL diluent yields 60mg/mL. 1,500mg vial: 10mL yields 150mg/mL.
2. CRITICAL: Use reconstituted glutathione within 24 hours. GSH oxidises rapidly once dissolved.
3. IV push: inject over minimum 2-3 minutes. Never rapid bolus.
4. IV drip: dilute in 100-250mL 0.9% NaCl. Infuse over 20-30 minutes.
5. IM: inject into deltoid or gluteal muscle with 23-25G x 1-1.5in needle.
6. No fasting required. Inject any time of day.
7. Protect reconstituted solution from light. Discard any unused portion after 24 hours.
8. Contraindicated in sulphite-sensitive asthma and known GSH allergy.
9. Pregnancy: GSH is endogenous but high-dose injectable use requires physician guidance.

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